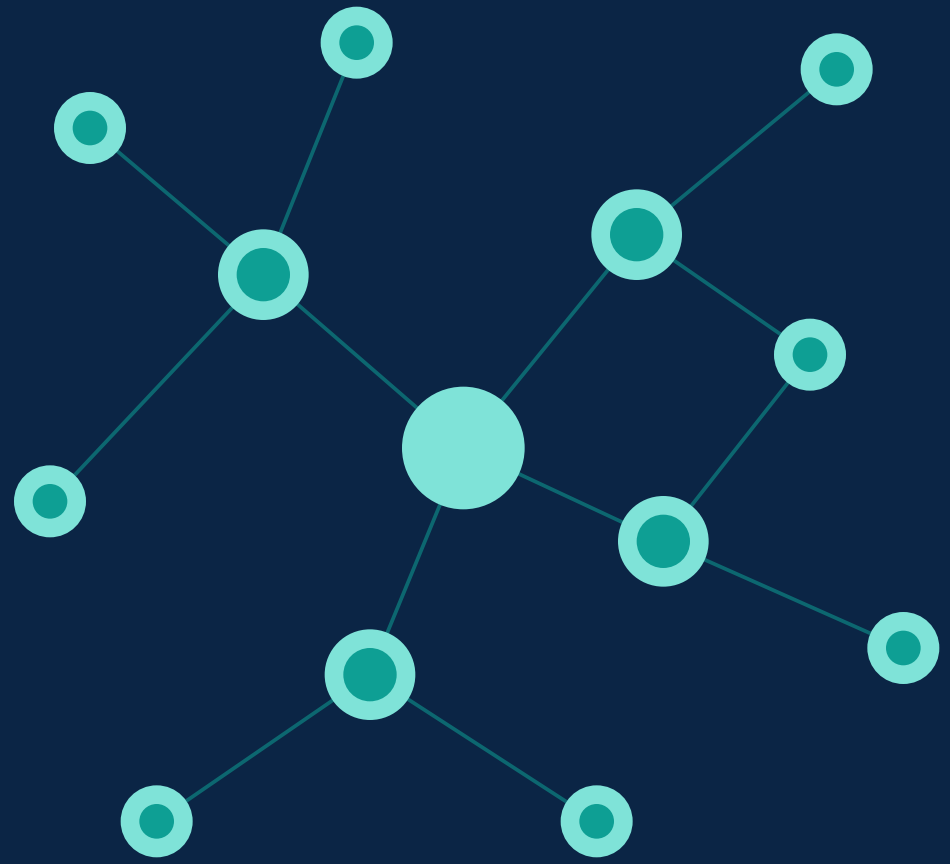
APPLICATION TO IIT MADRAS INCUBATION CELL

Payanar Type Script

The Business Knowledge Platform

Your business knowledge, written in a language AI understands.

Powered by **Payanarss Type Script (PTS)**: the open-source language and knowledge base behind every data model, rule and screen.



Bhuvaneswaran Shanmugam

Founder & Chief Architect

bhuvaness@gmail.com | payanarss.com | linkedin.com/in/bhuvaneswaran-shanmugam-b0672217

Phone: ADD PHONE NUMBER

A knowledge platform that makes software a right, not a privilege

PRODUCT	**Payanar**, the Business Knowledge Platform: a new kind of modular, AI-native ERP
COMPANY	CONFIRM LEGAL NAME (sources differ: Payanar Net Works / Payanarss Net Link / Viki Net Works)
INCORPORATION	CONFIRM: ENTITY, JURISDICTION, DATE
LANGUAGE	**PTS** = **P**ayanarss **T**ype **S**cript. Open-source; patent-pending
WEBSITE	payanarss.com
FOCUS MARKET	UAE first (SMBs and micro businesses); India-based roots and R&D
MENTORS / ADVISORS	ADD NAMES AND AFFILIATIONS

COMPANY VISION

“Software is a fundamental need for every business. It is not a privilege.”

WHAT IS PTS?

A language for the business itself

PTS describes a business’s data models, rules and screens in **one language that both software and AI can read**. It is stored as a searchable knowledge base and executed by our own interpreter, so the business is no longer buried in hand-written code and rigid tables.

- No single vendor owns your business logic: PTS is open for anyone to read, use and extend
- A change to the business is a change to PTS, not a rebuild of the core
- The right logic is delivered to each user from their own business configuration

An ERP team that discovered AI: 100+ years of ERP experience

BS

Bhuvaneswaran Shanmugam

Founder & Chief Architect

20+ years in enterprise software. Designed Payanarss Type Script (patent-pending) and built Payanar from the ground up. Published research on AI-native ERP architecture.

P

Mr. V. Padbanaban

Co-Founder

30+ years designing and implementing ERP for manufacturing industries across South India. Doctorate-level ERP architecture expertise.

LK

Dr. Lalit Khatri

Co-Founder & GCC Market Lead

30+ years in the UAE ERP field. Doctorate in marketing. End-to-end ERP life cycles, client onboarding and system integration.

SG

Mr. Srinivasan Govindarajan

Co-Founder & Manufacturing Lead

30+ years in manufacturing operations: workflows, production planning, inventory and quality management.

M

Mr. Manohar

Co-Founder & Finance Lead

25+ years in financial and business-process systems. Designed the Accounting Agent and financial-production integration logic.

PROPOSED SHAREHOLDING



Common ground: all five founders have strong manufacturing exposure and, per the IN5 deck, strong exposure to UAE real estate and logistics.

Development team: CONFIRM CURRENT SIZE AND LOCATION (IN5 deck says 50+ engineers, testers and PMs; please verify)

Today's ERP was built for large enterprises and keeps failing everyone else

Vendor lock-in

Organisations depend on one vendor. Switching or integrating alternatives is costly.

Data migration between ERPs

Moving ERPs means re-mapping data from one rigid data model to another. This is the main reason switching is so costly and risky.

Hard to customise

Even minor changes need deep core modifications, or a wait for the next release.

Integration challenges

Monolithic ERPs were not built for openness. Third-party integration is fragile.

Monolithic architecture

Giant, tightly coupled modules that depend on each other and break together.

High cost of change

Updating a deployed ERP is expensive, slow and risky.

THE RESULT FOR SMALL AND MID-SIZE BUSINESSES

50M	70%	65%	80%	18 mo / $2.5M	15%
SMBs globally have no ERP	cite “too expensive”	cite “too complex”	implementation failures (Gartner)	average ERP project	of SMBs use enterprise software

ADD PRIMARY SOURCES Statistics are carried over from the IN5 deck without citations. IITM asks for data to validate the problem, so please add the report names and years before submitting.

Our solution: one AI-readable language describes the whole business

PTS-driven, modular AI-agent ERP. PTS defines data models, business logic and UI behaviour, in one language.

1 Granular, layered agents

Functionality is split finely (“Employee Record”, “Onboarding”, “Training”). Each has UI, Business and Data layers and its own data store. No giant unified schema.

2 PTS: an AI-understandable language

All data models and business logic are defined in PTS, so the ERP is described in a form AI can read, not hand-coded per customer.

3 PTS Interpreter

We built the interpreter. It runs PTS directly, so adding or changing a data model or rule is a PTS change, not a rebuild of the core.

4 PTS as the AI knowledge base

PTS is stored in a Vector DB / Graph DB, so definitions are found by meaning and by how types, lookups and rules relate.

5 Configuration-driven UI and logic

PTS is selected from business configuration and lookup types. UI rendering and business logic both work from the knowledge base.

6 Security and governance

Each agent enforces authentication, authorisation and data privacy. Only authorised orchestrators can invoke sensitive agents.

Free to use, pay only for storage

No licence, no per-user, no per-module fees

Every use case on every device

Full mobile app for individuals; desktop for companies

Cross-company B2B

Company A's purchase order arrives in Company B's ERP as a sales order, via agent prompts

In plain words: write the business down once, and the software follows

THE ANALOGY

A traditional ERP is like a building poured in concrete. Moving one wall means breaking the structure, and only the original builder knows how.

Payanar is like a catalogue of labelled building blocks. The catalogue (PTS) describes every record, rule and screen. The software reads the catalogue and builds each business exactly what it chose. To change something, you edit the catalogue.

1 Describe

The business's data, rules and screens are written in PTS, which people and AI can both read.

2 Select

The business picks its configuration and lookup choices, for example the kind of business it runs.

3 Run

The matching PTS is found by meaning and delivered to the user's phone or computer.

TODAY	WITH PAYANAR
Logic lives in vendor-specific code and rigid tables	**Logic lives in PTS: open and readable by AI**
A change means core modification or waiting for a release	**A change is a PTS change**
Switching ERPs means re-mapping one rigid data model to another	**The business is described in a vendor-neutral form**
Menus and transaction codes	**Talk to the ERP: record a sale or send an invoice by voice**
Per-user and per-module licence fees	**Free app; pay only for the storage you choose**

The result: a change to the business is a change to PTS, not a rebuild of the core.

Customers: businesses that need software but were priced out of it

Micro businesses

1-9 employees

Solo traders, small shops and home-based businesses running on WhatsApp, Excel and paper. They need simple mobile software with no software cost to start.

Small and medium businesses

10-250 employees

Multi-staff firms that outgrew spreadsheets but find large ERP too complex and costly, with fees that grow per user and per module.

Real estate

First UAE market

Developers, property managers and agencies tracking units, tenants, leases, payments and maintenance across many properties and contracts.

Manufacturing

First UAE market

Factories and workshops managing production, inventory, quality, purchasing and finance. Our founders bring deep manufacturing experience.

WHO WRITES THE CHEQUE?

The owner of the micro business or SMB, who is also the user whose problem we solve. **The paying customer and the end customer are the same person.** They start free and pay only for the data storage they choose.

BEACHHEAD FOCUS

Application draft names **real estate and manufacturing** as first UAE markets; the IN5 deck names **real estate and logistics**. CONFIRM WHICH TO LEAD WITH

CUSTOMER VALIDATION

10 customers spoken to, to validate the problem and solution

From inception to date	ADD
Last 1 year	ADD
Last 6 months	ADD
Last 3 months	ADD

Feedback received: ADD NAME, AFFILIATION AND FEEDBACK FOR EACH OF THE 10 CONVERSATIONS

How it works: PTS becomes a knowledge base that delivers the right logic

1 Write the business in PTS

Data models, rules and screens are written as PTS: a typed tree of nodes (Id, ParentId, Name, Type, Attributes, Description). Our own **PTS Interpreter** runs it directly. Adding a model or rule is a PTS change, not a core rebuild.

2 Turn PTS into a knowledge base

PTS is stored in a **Vector DB and a Graph DB**. The vector side finds definitions by meaning; the graph side holds how types, lookups and rules relate. AI navigates this pre-authored knowledge rather than inventing logic.

3 Deliver it per business

The business's **configuration and lookup selections** choose the PTS it needs. The matching logic and UI definitions are downloaded to the user's mobile or desktop. Change a lookup and the rules and screens change with it.

THE AGENT LAYER

Granular agents, each with its own store

UI layer	Business layer	Data layer
rendered from PTS	rules from PTS	own data store

Prompt-based agent-to-agent communication
Open REST APIs (OpenAPI / Swagger)
Per-agent authentication and authorisation
Comprehensive logging and monitoring

ADD Schematic or screenshots (IITM asks for these): PTS tree view, mobile app, Viki voice assistant, sample PTS definition.

Why this technology is needed now, and why it is feasible

WHY IT IS NEEDED

- **The ERP model is failing.** Logic hidden in vendor code and rigid tables makes change slow, costly and risky.
- **Most of the market is left out.** Smaller businesses cannot afford or manage traditional ERP.
- **Menus are a barrier.** People want to talk to the system, not learn transaction codes.

WHY NOW: THREE ENABLING TECHNOLOGIES

Generative AI and agents

Business logic can be invoked by simple prompts.

Composable microservices

Cloud-native, API-first ERP split into small agents.

Prompt-driven interoperability

Replaces brittle point-to-point B2B links.

WHY WE THINK IT IS TECHNICALLY FEASIBLE

BUILT

The PTS interpreter exists and runs PTS directly.

BUILT

A PTS knowledge base of 13,600+ typed nodes CONFIRM CURRENT COUNT

DESIGNED FOR SAFETY

AI retrieves pre-authored definitions and returns structured intent; it does not generate business logic.

PUBLISHED

SSRN white paper on AI-driven ERP architecture.

OPEN STANDARDS

Every agent exposes REST APIs documented in OpenAPI / Swagger.

PROTECTED

PTS is patent-pending ADD FILING DETAILS

Existing solutions keep logic in vendor tooling; Payanar keeps it in open knowledge

	Microsoft Dynamics 365	Oracle NetSuite	SAP Business One	Zoho	Payanar
Where business logic lives	Dataverse, X++, Power Platform	SuiteScript, SuiteFlow	SAP SDK, user-defined fields, add-ons	Deluge (Zoho scripting)	**PTS: an open-source knowledge base AI can read**
How you change it	Vendor-specific skills and tools	Vendor-specific skills and tools	Vendor-specific skills and tools	Vendor-specific skills and tools	**Select configuration and lookups; a change is a PTS change**
Aimed at	Established suite	Mid-market cloud ERP	Small and mid-size firms	ADD	**Micro businesses to companies; mobile for everyone**
What the customer pays	Licence-based models where fees grow with users and modules ADD VENDOR PRICING				**Free app; storage only: AED 500 or 1,000 / month**

UNIQUE IN TECHNOLOGY

- Business knowledge as open PTS, not code and tables
- AI-readable knowledge base (Vector + Graph DB)
- Granular agents with an agent marketplace: no vendor lock-in, no downtime to add a function

UNIQUE IN BUSINESS MODEL

- Free to download and use: no licence, per-user or per-module fees
- Revenue grows only as customers store more data
- Proof before payment: we run your workflow first, one department at a time

A large, underserved market, and we start where it is closest: the UAE

50M
SMBs globally have no ERP

$21-85B
Total addressable market

18-22%
MENA market CAGR

<5%
Current MENA ERP penetration

WHY THE UAE

- SMEs are **94.4%** of all UAE companies and **63.5%** of non-oil GDP (2022)
- **15%+** annual SMB growth, backed by UAE Vision 2031 and the National AI Strategy

UAE: FROM MARKET TO FIVE-YEAR PLAN

557,000
SMEs across the UAE

170K - 220K
Tier-2 and Tier-3 SMEs without enterprise-grade ERP: our addressable market

~1,825
clients needed by Year 5: about 1% of Tier-2/3 SMEs

Bars are illustrative, not to scale. Five-year plan: ~1,825 clients, AED 10.95M a year at AED 500 a month (AED 19.7M a year at AED 1,000 a month).

ADD SOURCES Market figures are carried over from the IN5 deck without citations. The $21-85B range is wide; add the method and reports behind it, and the basis for the 170K-220K Tier-2/3 estimate.

Differentiation, sustainability and entry barriers

What is different

- **Technology:** business knowledge in open PTS, run by our own interpreter and searchable by AI
- **Business model:** free software; pay only for data storage
- **Reach:** every use case on mobile, so micro businesses are served, not just companies

How we stay ahead

- A growing PTS knowledge base: every sector and use case encoded makes the next faster to deliver
- Patent-pending PTS, with an open ecosystem and agent marketplace that invites third-party agents
- Founder depth: 100+ years of ERP and manufacturing experience, 30+ years in the UAE market

Entry barriers for others

- ERP domain depth is needed to encode real business knowledge correctly
- A broad, trusted knowledge base takes time to build
- Security, authorisation and data-governance expectations for business data
- Incumbents are tied to their own languages and platforms

REVIEW The IITM form asks for competitor size (revenue, customers). Add sizes with sources for Microsoft, Oracle, SAP and Zoho. The “stay ahead” and “entry barrier” points are drafted from your IN5 content, so please confirm they reflect your view.

Go-to-market: start free, prove value, then grow with the customer's data

Download free → **Use the application** → **Run one department as proof** → **Choose a storage plan, pay for storage**

DISTRIBUTION CHANNEL

- Free download on mobile and desktop
- Content marketing, LinkedIn, Product Hunt, webinars and training
- Founder network in the UAE (30+ years in UAE ERP)
- Agent marketplace over time

KEY PARTNERS

- Incubators and accelerators: IITM Incubation Cell, in5, Hub71, MBRIF (submissions made)
- Pilot partners (discussions in progress)
- Open-source and agent communities
- Cloud and vector-DB infrastructure providers

KEY ACTIVITIES

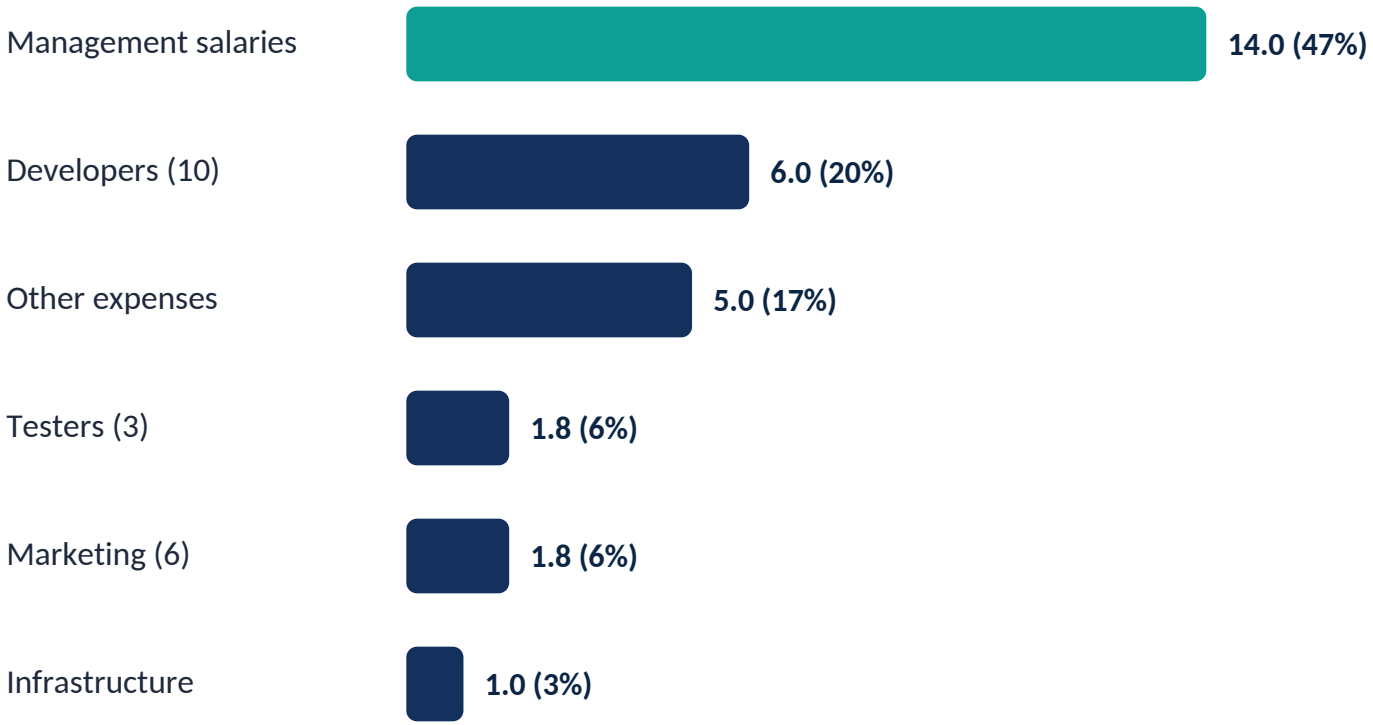
- Run customer workflows as proof, one department at a time
- Complete modules: Inventory + Sales, then Finance + Procurement
- Release the browser tool v1
- Win pilot customers; encode sectors into PTS

KEY RESOURCES

- PTS, the interpreter and the knowledge base
- Five founders with 100+ years of ERP experience
- Engineering team CONFIRM SIZE
- Planned UAE team: AI, partnerships, sales (Abu Dhabi / Dubai)

Cost structure: people are the main investment

FIVE-YEAR EXPENSES, UAE PLAN (AED MILLION)



Total, five years: AED 29.6M

WHERE WE SPEND MOST

~74%

of five-year cost is management, developer and tester salaries (AED 21.8M of 29.6M).

SEED-STAGE SPEND (AED 2.4-2.6M)

- 50% product and AI development
- 20% infrastructure and cloud
- 15% UAE setup and operations
- 10% business development
- 5% marketing

Detailed on the funding slide.

Revenue model: the app is free; customers pay only for the storage they choose

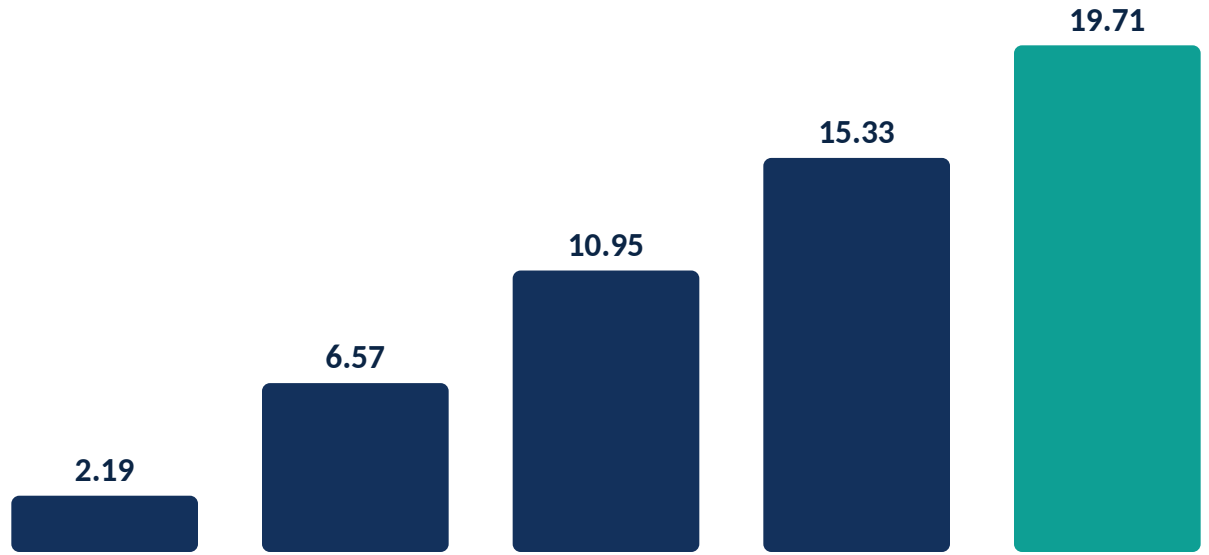
PRICING

Application: FREE. No licence, no subscription, no per-user or per-module fees.

50 GB	100 GB
AED 500	AED 1,000
per month	per month
storage plan	storage plan

- **Zero barrier to start:** free download
- **Proof before payment:** we run your workflow first
- **Revenue grows with you:** each module you add stores more data

FIVE-YEAR REVENUE PROJECTION, UAE ONLY (AED MILLION)



AED 54.75M	AED 25.15M	Year 1.4	Year 2.7
5-year revenue	EBITDA (45.9%)	monthly break-even	cumulative break-even

Founder planning assumptions, not achieved results: UAE only; 365 new clients a year; all clients on the 100 GB plan (AED 1,000 a month); no churn; costs flat at plan.

Where we are: proof of concept built, research published, validation under way

PTS and PTS Interpreter	PTS defined; interpreter built and runs PTS directly	BUILT
PTS knowledge base	13,600+ typed nodes in a recursive tree CONFIRM COUNT	BUILT
Product prototype	Employee Agent prototype, in development per IN5 deck. UPDATE: MODULES, MVP STATUS, SCREENSHOTS	IN PROGRESS
Research	MAA ERP white paper on SSRN (published). CitizenssApp paper submitted. CONFIRM STATUS	PUBLISHED
Customer validation	10 customers spoken to; pilot-partner discussions in progress. ADD PILOT STATUS	IN PROGRESS
External institutes and industry	Incubator submissions: in5, Hub71, MBRIF. IITM involvement to date: CONFIRM	SUBMITTED
Experts and mentors consulted	ADD NAMES AND AFFILIATIONS	TO ADD
Patent	PTS is patent-pending. ADD APPLICATION NO., DATE, JURISDICTION	PENDING

Stage: proof of concept moving into pilot validation. CONFIRM ANY PAYING CUSTOMERS OR REVENUE TO DATE

Next 12 months: from modules and pilots to paying customers

MONTHS 1-3 Build

- Inventory + Sales module complete
- Browser tool v1 released
- Pilot onboarding begins

MONTHS 4-6 Pilot

- 5 pilot customers, one department each
- Finance + Procurement modules
- Seed round conversations after incubator approval

MONTHS 7-9 Convert

- Pilots move to paid storage plans
- UAE entity and workspace set up
- First sales and customer-success hires

MONTHS 10-12 Grow

- Target: 20 paying UAE SMBs
- Agent marketplace groundwork
- Review pricing and storage plans with real usage

RESOURCES NEEDED

2 senior AI / Vector-DB engineers, 1 frontend engineer, 1-2 sales hires, 1 customer-success manager, cloud and vector-DB infrastructure, UAE entity and workspace.

CONFIRM The IN5 roadmap was dated Q1-Q3 2025 and is out of date, so milestones are re-based as months 1-12 from your IN5 plan. The IITM form also asks for financial projections per milestone; please add a per-phase budget.

Funding: AED 2.4-2.6M seed, raised on proof rather than promise

THE ASK

AED 2.4-2.6M

seed round to launch Payanar in the UAE

Runway: 18-24 months

HISTORY OF FUNDS RAISED

No external investment raised. We have not approached investors, by design: we are validating with customers first, and will approach investors once an incubator approves us, with that validation behind us.

CONFIRM any founder or self-funded investment to date

USE OF FUNDS

Product and AI development — 50% | AED 1.19-1.29M

2 senior AI / Vector-DB engineers, 1 frontend engineer, Finance and Procurement modules, browser tool v1

Infrastructure and cloud — 20% | AED 0.48-0.51M

Cloud hosting, production vector DB, security and compliance, CI/CD

UAE setup and operations — 15% | AED 0.36-0.39M

Workspace (1 year), legal and compliance, UAE entity, administration

Business development — 10% | AED 0.24-0.26M

Sales (1-2), customer success, pilot acquisition, partnerships

Marketing — 5% | AED 0.12-0.13M

Content, LinkedIn and Product Hunt, webinars, brand identity

What we ask of IIT Madras and the IITM Incubation Cell

Incubation and workspace

- Incubation space and facilities for the core team
- A base for customer meetings, demos and technical work

Technical and expert mentorship

- Review of the PTS design and the knowledge-base approach
- Mentors in enterprise software, AI and SaaS pricing
- Feedback on our free-software, pay-for-storage model

Customer and industry access

- Introductions to SMBs and industry partners for pilots
- Goal: 20 paying SMBs on Payanar

IP and patent support

- Guidance on strengthening and managing the PTS patent
- Advice on pairing an open-source language with protected IP

Investor network

- Introductions to seed investors for the AED 2.4-2.6M round
- Pitch feedback before investor meetings

OUR COMMITMENT

We will use the programme to turn a working proof of concept into paying customers, and to make PTS a credible open standard for business knowledge.

CONFIRM These asks are adapted from your IN5 asks. Please tailor them to what IITM Incubation Cell actually offers, and add your reason for choosing IITM (for example, research depth, India base and R&D, alumni network).

Applicant details

PRIMARY APPLICANT

Bhuvaneswaran Shanmugam

Founder & Chief Architect

EMAIL	bhuvaness@gmail.com
LINKEDIN	linkedin.com/in/bhuvaneswaran-shanmugam-b0672217
RESEARCH	SSRN: papers.ssrn.com/sol3/papers.cfm?abstract_id=5432994
PHONE	ADD
ADDRESS	ADD COMPLETE CONTACT ADDRESS

BACKGROUND

- 20+ years in enterprise IT: cloud architecture (AWS, PCF, Azure), microservices, AI-native systems
- BSc Mathematics, Bharathidasan University
- Published SSRN research on AI-driven ERP architecture

CO-FOUNDERS AND PROMOTERS

Mr. V. Padbanaban	ADD CONTACT DETAILS
Dr. Lalit Khatri	ADD CONTACT DETAILS
Mr. Srinivasan Govindarajan	ADD CONTACT DETAILS
Mr. Manohar	ADD CONTACT DETAILS

Company promoters / directors: ADD, PER INCORPORATION DOCUMENTS

One-page CV for each person: ATTACH

“Digital empowerment is no longer a privilege. It is a fundamental need for every citizen. CRUD is the new pen and paper.”

Software is not a privilege. Help us make it available to every business.